JOBSHEET 11

LINKED LIST

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KELAS : TI 1-G

NO. ABSEN : 18

Percobaan 1 (Pembuatan Single Linked List).

Code Mahasiswa18.java:

```java
package Jobsheet11;

public class Mahasiswa18 {
    String nim;
    String nama;
    String kelas;
    double ipk;

    public Mahasiswa18() {
    }

    public Mahasiswa18(String nim, String nama, String kelas, double ipk) {
        this.nim = nim;
        this.nama = nama;
        this.kelas = kelas;
        this.ipk = ipk;
    }

    public void tampilInformasi() {
        System.out.println("NIM    : " + nim);
        System.out.println("Nama   : " + nama);
        System.out.println("Kelas  : " + kelas);
        System.out.println("IPK    : " + ipk);
        System.out.println("------------------------");
    }
}
```

Code Node18.java:

```java
package Jobsheet11;

public class Node18 {
    Mahasiswa18 data;
    Node18 next;

    public Node18(Mahasiswa18 data, Node18 next) {
        this.data = data;
        this.next = next;
    }
}

```

Code SingleLinkedList18.java:

```java
package Jobsheet11;

public class SingleLinkedList18 {
    Node18 head;
    Node18 tail;

    boolean isEmpty() {
        return (head == null);
    }

    public void print() {
        if (!isEmpty()) {
            Node18 tmp = head;
            System.out.println("Isi Linked List:");
            while (tmp != null) {
                tmp.data.tampilInformasi();
                tmp = tmp.next;
            }
        } else {
            System.out.println("Linked list kosong");
        }
        System.out.println();
    }

    public void addFirst(Mahasiswa18 input) {
        Node18 ndInput = new Node18(input, null);
        if (isEmpty()) {
            head = ndInput;
            tail = ndInput;
        } else {
            ndInput.next = head;
            head = ndInput;
        }
    }

    public void addLast(Mahasiswa18 input) {
        Node18 ndInput = new Node18(input, null);
        if (isEmpty()) {
            head = ndInput;
            tail = ndInput;
        } else {
            tail.next = ndInput;
            tail = ndInput;
        }
    }

    public void insertAfter(String key, Mahasiswa18 input) {
        Node18 ndInput = new Node18(input, null);
        Node18 temp = head;
        do {
            if (temp.data.nama.equalsIgnoreCase(key)) {
                ndInput.next = temp.next;
                temp.next = ndInput;
                if (ndInput.next == null) {
                    tail = ndInput;
                }
                break;
            }
            temp = temp.next;
        } while (temp != null);
    }

    public void insertAt(int index, Mahasiswa18 input) {
        if (index < 0) {
            System.out.println("indeks salah");
        } else if (index == 0) {
            addFirst(input);
        } else {
            Node18 temp = head;
            for (int i = 0; i < index - 1; i++) {
                temp = temp.next;
            }

            temp.next = new Node18(input, temp.next);

            if (temp.next.next == null) {
                tail = temp.next;
            }
        }
    }
}
```

Code SLLMain18.java:

```java
package Jobsheet11;

public class SLLMain18 {
    public static void main(String[] args) {

        SingleLinkedList18 sll = new SingleLinkedList18();

        Mahasiswa18 mhs1 = new Mahasiswa18("001", "Andi", "TI-1A", 3.5);
        Mahasiswa18 mhs2 = new Mahasiswa18("002", "Budi", "TI-1B", 3.7);
        Mahasiswa18 mhs3 = new Mahasiswa18("003", "Dirga", "TI-1C", 3.8);
        Mahasiswa18 mhs4 = new Mahasiswa18("004", "Citra", "TI-1D", 3.6);

        sll.print();

        sll.addFirst(mhs4);
        sll.print();

        sll.addLast(mhs1);
        sll.print();

        sll.insertAfter("Dirga", mhs3);
        sll.print();

        sll.insertAt(2, mhs2);
        sll.print();
    }
}

```

Outputnya:

```
Linked list kosong

Isi Linked List:
Citra           004             TI-1D       3,60

Isi Linked List:
Citra           004             TI-1D       3,60
Andi            001             TI-1A       3,50

Isi Linked List:
Citra           004             TI-1D       3,60
Andi            001             TI-1A       3,50

Isi Linked List:
Citra           004             TI-1D       3,60
Andi            001             TI-1A       3,50
Budi            002             TI-1B       3,70
```

Pertanyaan!

1. Output tersebut muncul karena pada saat print() pertama kali dipanggil, linked list masih belum berisi data. Nilai head masih null, sehingga method isEmpty() mengembalikan nilai true. Sehingga, program menampilkan pesan bahwa linked list kosong.
2. Variabel temp digunakan sebagai penunjuk sementara untuk menelusuri isi linked list dari awal sampai akhir. Dengan menggunakan temp, proses traversal bisa dilakukan tanpa mengubah posisi head, sehingga struktur data tetap aman dan tidak rusak.
3. Codenya:

```java
package Jobsheet11;

import java.util.Scanner;

public class SLLMain18 {
    public static void main(String[] args) {

        SingleLinkedList18 sll = new SingleLinkedList18();

        Scanner sc = new Scanner(System.in);

        boolean lanjut = true;
        while (lanjut) {
            System.out.println("Masukkan data mahasiswa:");
            System.out.print("Nama: ");
            String nama = sc.nextLine();
            System.out.print("NIM: ");
            String nim = sc.nextLine();
            System.out.print("Kelas: ");
            String kelas = sc.nextLine();
            System.out.print("IPK: ");
            double ipk = sc.nextDouble();
            sc.nextLine();

            Mahasiswa18 mhs = new Mahasiswa18(nama, nim, kelas, ipk);

            System.out.println("\nPilih operasi penambahan:");
            System.out.println("1. Tambah di awal (addFirst)");
            System.out.println("2. Tambah di akhir (addLast)");
            System.out.println("3. Tambah di indeks tertentu (insertAt)");
            System.out.println("4. Tambah setelah nama tertentu (insertAfter)");
            System.out.print("Pilih: ");
            int pilihan = sc.nextInt();
            sc.nextLine();

            switch (pilihan) {
                case 1:
                    sll.addFirst(mhs);
                    break;
                case 2:
                    sll.addLast(mhs);
                    break;
                case 3:
                    System.out.print("Masukkan indeks: ");
                    int index = sc.nextInt();
                    sc.nextLine();
                    sll.insertAt(index, mhs);
                    break;
                case 4:
                    System.out.print("Masukkan nama yang mendahului: ");
                    String key = sc.nextLine();
                    sll.insertAfter(key, mhs);
                    break;
                default:
                    System.out.println("Pilihan tidak valid!");
            }

            sll.print();

            System.out.print("\nApakah Anda ingin menambah data lagi? (y/n): ");
            String jawab = sc.nextLine();
            if (jawab.equalsIgnoreCase("n")) {
                lanjut = false;
            }
        }

        System.out.println("Program selesai.");

        sc.close();
    }
}
```

```
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData> & "C:\Program Files\Java\jdk-24\bin\java.exe" "-XX:+ShowCodeDetailsInExceptionMessages" "-cp" "C:\Users\user\AppData\Roaming\Code\User\workspaceStorage\536fd7849
Masukkan data mahasiswa:
Nama:
```

Percobaan 2 (Modifikasi Elemen pada Single Linked List).

Code SingleLinkedList18.java:

Jobsheet11 > SingleLinkedList18.java > SingleLinkedList18 > removeAt(int)

```java
public class SingleLinkedList18 {

    public void getData(int index) {
        Node18 tmp = head;
        for (int i = 0; i < index; i++) {
            tmp = tmp.next;
        }
        tmp.data.tampilInformasi();
    }

    public int indexOf(String key) {
        Node18 tmp = head;
        int index = 0;
        while (tmp != null && !tmp.data.nama.equalsIgnoreCase(key)) {
            tmp = tmp.next;
            index++;
        }

        if (tmp == null) {
            return -1;
        } else {
            return index;
        }
    }

    public void removeFirst() {
        if (isEmpty()) {
            System.out.println(x: "Linked List masih Kosong, tidak dapat dihapus!");
        } else if (head == tail) {
            head = tail = null;
        } else {
            head = head.next;
        }
    }

    public void removeLast() {
        if (isEmpty()) {
            System.out.println(x: "Linked List masih Kosong, tidak dapat dihapus!");
        } else if (head == tail) {
            head = tail = null;
        } else {
            Node18 temp = head;
            while (temp.next != tail) {
                temp = temp.next;
            }
            temp.next = null;
            tail = temp;
        }
    }

    public void remove(String key) {
        if (isEmpty()) {
            System.out.println(x: "Linked List masih Kosong, tidak dapat dihapus!");
        } else {
            Node18 temp = head;
            while (temp != null) {
                if ((temp.data.nama.equalsIgnoreCase(key)) && (temp == head)) {
                    this.removeFirst();
                    break;
                } else if (temp.data.nama.equalsIgnoreCase(key)) {
                    temp.next = temp.next.next;
                    if (temp.next == null) {
                        tail = temp;
                    }
                    break;
                }
                temp = temp.next;
            }
        }
    }

    public void removeAt(int index) {
        if (index == 0) {
            removeFirst();
        } else {
            Node18 temp = head;
            for (int i = 0; i < index - 1; i++) {
                temp = temp.next;
            }
            temp.next = temp.next.next;
            if (temp.next == null) {
                tail = temp;
            }
        }
    }
```

Code SLLMain18.java:

```java
package Jobsheet11;

public class SLLMain18 {
    public static void main(String[] args) {

        SingleLinkedList18 sll = new SingleLinkedList18();

        Mahasiswa18 mhs1 = new Mahasiswa18("001", "Alvaro", "1A", 4.0);
        Mahasiswa18 mhs2 = new Mahasiswa18("002", "Bimon", "2B", 3.8);
        Mahasiswa18 mhs3 = new Mahasiswa18("003", "Cintia", "3C", 3.5);
        Mahasiswa18 mhs4 = new Mahasiswa18("004", "Dirga", "4D", 3.6);

        sll.print();

        sll.addFirst(mhs4);
        sll.print();

        sll.addLast(mhs1);
        sll.print();

        sll.insertAfter("Dirga", mhs3);

        sll.insertAt(2, mhs2);
        sll.print();

        System.out.println("Data index 1:");
        sll.getData(1);

        System.out.println("\nData mahasiswa an Bimon berada pada index : " + sll.indexOf("Bimon"));
        System.out.println();

        sll.removeFirst();
        sll.removeLast();
        sll.print();

        sll.removeAt(0);
        sll.print();
    }
}
```

Outputnya:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    GITLENS

itmaStrukturData_b46f464f\bin' 'Jobsheet11.SLLMain18'
Linked list kosong

Isi Linked List:

Dirga          004            4D        3,6
Isi Linked List:

Dirga          004            4D        3,6
Alvaro         001            1A        4,0
Isi Linked List:

Dirga          004            4D        3,6
Cintia         003            3C        3,5
Bimon          002            2B        3,8
Alvaro         001            1A        4,0
Data index 1:

Cintia         003            3C        3,5
Data mahasiswa an Bimon berada pada index : 2

Isi Linked List:

Cintia         003            3C        3,5
Bimon          002            2B        3,8
Isi Linked List:

Bimon          002            2B        3,8
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```

Pertanyaan!

1. Untuk menghentikan perulangan ketika data sudah penuh.
2. Kode temp.next = temp.next.next; menghapus node dengan melewati node tersebut, sedangkan if (temp.next == null) memastikan jika yang dihapus adalah node terakhir, maka tail diperbarui agar tetap menunjuk ke node terakhir yang benar.

Tugas!

Code MahasiswaAntrian18:

```java
package Jobsheet11.TugasJobsheet11;

public class MahasiswaAntrian18 {
    String nim, nama, keperluan;

    public MahasiswaAntrian18(String nim, String nama, String keperluan) {
        this.nim = nim;
        this.nama = nama;
        this.keperluan = keperluan;
    }

    public void tampil() {
        System.out.println(nim + " | " + nama + " | " + keperluan);
    }
}
```

Code NodeAntrian18:

```java
package Jobsheet11.TugasJobsheet11;

public class NodeAntrian18 {
    MahasiswaAntrian18 data;
    NodeAntrian18 next;

    public NodeAntrian18(MahasiswaAntrian18 data) {
        this.data = data;
        this.next = null;
    }
}
```

Code QueueLinkedList18:

```java
package Jobsheet11.TugasJobsheet11;

public class QueueLinkedList18 {
    NodeAntrian18 front, rear;
    int size = 0;
    int max = 10;

    boolean isEmpty() {
        return front == null;
    }

    boolean isFull() {
        return size == max;
    }

    void clear() {
        front = rear = null;
        size = 0;
        System.out.println("Antrian dikosongkan");
    }

    void enqueue(MahasiswaAntrian18 mhs) {
        if (isFull()) {
            System.out.println("Antrian penuh!");
            return;
        }

        NodeAntrian18 newNode = new NodeAntrian18(mhs);

        if (isEmpty()) {
            front = rear = newNode;
        } else {
            rear.next = newNode;
            rear = newNode;
        }
        size++;
        System.out.println(mhs.nama + " berhasil masuk antrian");
    }

    void dequeue() {
        if (isEmpty()) {
            System.out.println("Antrian kosong!");
            return;
        }

        System.out.print("Memanggil: ");
        front.data.tampil();

        front = front.next;
        size--;

        if (front == null) {
            rear = null;
        }
    }

    void peek() {
        if (!isEmpty()) {
            System.out.print("Antrian terdepan: ");
            front.data.tampil();

            System.out.print("Antrian terakhir: ");
            rear.data.tampil();
        } else {
            System.out.println("Antrian kosong!");
        }
    }

    void print() {
        if (isEmpty()) {
            System.out.println("Antrian kosong!");
            return;
        }

        NodeAntrian18 temp = front;
        System.out.println("Daftar Antrian:");
        while (temp != null) {
            temp.data.tampil();
            temp = temp.next;
        }
    }

    void jumlah() {
        System.out.println("Jumlah antrian: " + size);
    }
}
```

Code MainQueue18:

```java
package Jobsheet11.TugasJobsheet11;

import java.util.Scanner;

public class MainQueue18 {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        QueueLinkedList18 q = new QueueLinkedList18();

        int pilih;

        do {
            System.out.println(x: "\n=== ANTRIAN KEMAHASISWAAN ===");
            System.out.println(x: "1. Tambah Antrian");
            System.out.println(x: "2. Panggil Antrian");
            System.out.println(x: "3. Cek Antrian Kosong");
            System.out.println(x: "4. Cek Antrian Penuh");
            System.out.println(x: "5. Kosongkan Antrian");
            System.out.println(x: "6. Lihat Depan & Belakang");
            System.out.println(x: "7. Tampilkan Semua");
            System.out.println(x: "8. Jumlah Antrian");
            System.out.println(x: "0. Keluar");
            System.out.print(s: "Pilih: ");
            pilih = sc.nextInt();
            sc.nextLine();

            switch (pilih) {
                case 1:
                    System.out.print(s: "NIM: ");
                    String nim = sc.nextLine();
                    System.out.print(s: "Nama: ");
                    String nama = sc.nextLine();
                    System.out.print(s: "Keperluan: ");
                    String kep = sc.nextLine();

                    q.enqueue(new MahasiswaAntrian18(nim, nama, kep));
                    break;

                case 2:
                    q.dequeue();
                    break;

                case 3:
                    System.out.println(q.isEmpty() ? "Kosong" : "Tidak kosong");
                    break;

                case 4:
                    System.out.println(q.isFull() ? "Penuh" : "Tidak penuh");
                    break;

                case 5:
                    q.clear();
                    break;

                case 6:
                    q.peek();
                    break;

                case 7:
                    q.print();
                    break;

                case 8:
                    q.jumlah();
                    break;
            }

        } while (pilih != 0);

        sc.close();
    }
}
```

Outputnya:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    GITLENS

0. Keluar
Pilih: 3
Tidak kosong

=== ANTRIAN KEMAHASISWAAN ===
1. Tambah Antrian
2. Panggil Antrian
3. Cek Antrian Kosong
4. Cek Antrian Penuh
5. Kosongkan Antrian
6. Lihat Depan & Belakang
7. Tampilkan Semua
8. Jumlah Antrian
0. Keluar
Pilih: 4
Tidak penuh

=== ANTRIAN KEMAHASISWAAN ===
1. Tambah Antrian
2. Panggil Antrian
3. Cek Antrian Kosong
4. Cek Antrian Penuh
5. Kosongkan Antrian
6. Lihat Depan & Belakang
7. Tampilkan Semua
8. Jumlah Antrian
0. Keluar
Pilih: 8
Jumlah antrian: 2

=== ANTRIAN KEMAHASISWAAN ===
1. Tambah Antrian
2. Panggil Antrian
3. Cek Antrian Kosong
4. Cek Antrian Penuh
5. Kosongkan Antrian
6. Lihat Depan & Belakang
7. Tampilkan Semua
8. Jumlah Antrian
0. Keluar
Pilih: 7
Daftar Antrian:
11 | Vir | CEK
22 | Ben | CEK

=== ANTRIAN KEMAHASISWAAN ===
1. Tambah Antrian
2. Panggil Antrian
3. Cek Antrian Kosong
4. Cek Antrian Penuh
5. Kosongkan Antrian
6. Lihat Depan & Belakang
7. Tampilkan Semua
8. Jumlah Antrian
0. Keluar
Pilih: 0
PS C:\TI\ASD\PraktikumAlgoritmaStrukturData>
```